



SMART INDIA HACKATHON 2026



PATH PLANNING AND COLLISION AVOIDANCE

- **Problem Statement ID – SIH26037**
- **Problem Statement Title-**
Adaptive Path Planning and Collision Avoidance
for Autonomous Vehicles on Unstructured Indian Roads
- **Theme- Robotics and Drones**
- **PS Category- Software**
- **Team ID- SIH26-A0H-T333**
- **Team Name (Registered on portal) : Young Innovators**





PROPOSED SOLUTION

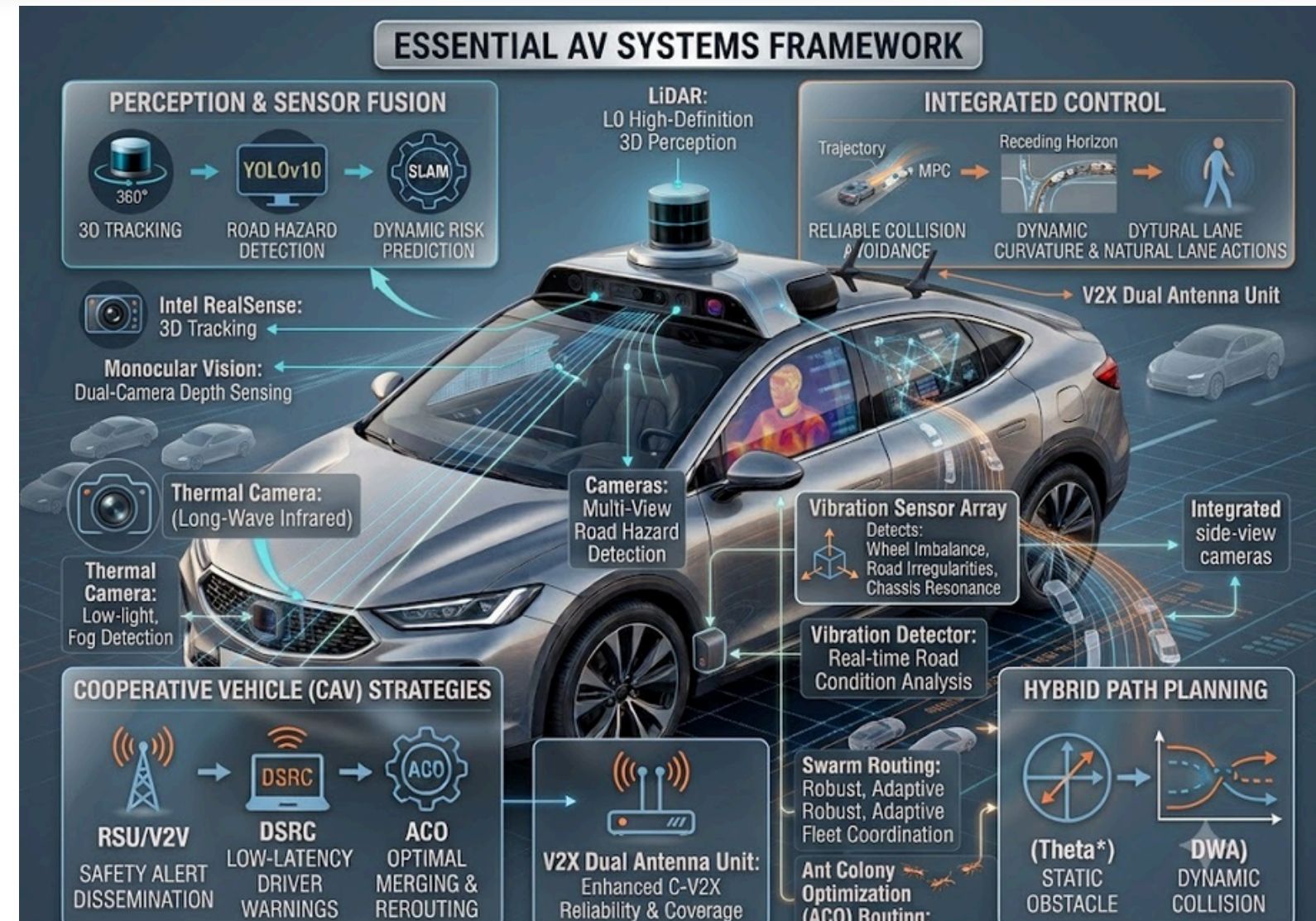


MOTIVATION TO ADDRESS THE PROBLEM AND GENERATE SOLUTION

- **Fatal wrong-side collisions:** According to **Ministry of Road Transport and Highways (MoRTH)** statistics, driving on the wrong side of the road accounts for roughly 5.5% of total road-use violations, leading to abrupt, high-fatality head-on crashes
- **High-velocity over-speeding crashes:** MoRTH data confirms over-speeding is the primary killer in vehicle-to-vehicle impacts, responsible for **70.3% of all recorded road fatalities** due to cabin structural collapse.

WHY WE STAND OUT

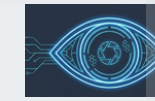
1. Built **for India, not adapted** for India.
2. Most ADAS assumes clean lanes and structured traffic, while this is designed from the ground up for mixed vehicles, no lane markings, monsoon glare, and stray animals.
4. **10-16x cheaper** than commercial systems. **₹90,000 hardware** cost vs ₹9.5L-15L+ for existing Level-2+ ADAS — makes safety tech viable for mass-market Indian vehicles, not just luxury cars.
5. Faster than human reflexes. **Sub-5ms emergency response** is a hard real-time, safety-certified capability.
6. Industrially credible, Built natively on MATLAB/Simulink with ISO 26262 ASIL-D compliance and AUTOSAR deployment



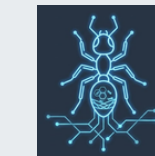
OUR REVOLUTIONARY SOLUTION

1. **Fuses RGB-D camera + 2D LiDAR + thermal + vibration sensors** for full-scene awareness, day or night
2. **Bio-inspired vision** corrects for rain, glare, and waterlogged potholes in real time
3. Detects **free driveable space** directly — no dependency on painted lane markings
4. Tracks each object across frames and **predicts its next few seconds of motion**
5. **Flags in advance** whether an object's path will actually conflict with the vehicle's path
6. **Continuously scores every object** LOW / MEDIUM / HIGH risk using distance, speed, and collision timing
7. Plans **smooth, safe paths** around both static and moving obstacles simultaneously
8. **Auto-selects the best trajectory** by balancing safety, comfort, and path efficiency
9. A **dedicated emergency-reflex layer** can brake or steer away from sudden danger in under 5 milliseconds
10. Vehicles **share real-time hazard alerts** with each other and nearby infrastructure
11. Fully **tested in digital-twin simulation** and hardware-in-loop rigs before deployment

KEY FEATURES



- **Peripheral Spatial Mapping:** Synthesizes "invisible lanes" on unmapped roads (Chameleon-Eye Model).



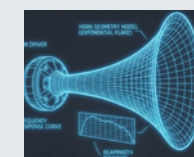
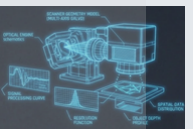
Colony Route Finding: micro-trajectories through chaotic roads (Ant Colony Optimization).

Visual Expansion Emergency Brakes: Instant safety stops in <5ms (Lobula Giant Movement Detector Neuron).



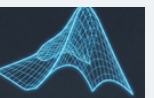
Multi-Agent Collision Avoidance: separation, alignment, and cohesion rules to flow through traffic (Reynolds' Boids Swarm Model).

Thermal & Refraction Pothole Scanner: LWIR neutralizes glare, while Snell's refraction math measures puddle-hidden pothole depth.



Context-Aware Acoustic Horn Alerts: Emits targeted horn pulses based on collision vectors to maintain traffic flow.

MATLAB & RoadRunner Validation: Closed-loop verified across 5 core Indian scenarios (Village, Intersection, Merge, Market, Cattle Crossing).





FEASIBILITY AND VIABILITY



Commercial & Market Viability

- **Democratizing Level 2+ Autonomy:** Replaces expensive, premium-only ADAS suites with an optimized, low-cost bio-inspired configuration to bring advanced autonomy to mass-market sub-compact vehicles.
- **Low-Cost Disruptive BOM:** Eliminates costly 3D LiDAR arrays using a streamlined, high-efficiency sensor footprint:
- **Stereo Depth Camera (RGB-D):** Handles primary visual identification without complex multi-camera clusters.
- **2D Solid-State LiDAR:** Delivers localized planar scanning in place of bulky 360° mechanical LiDARs.
- **CAN-Data & IMU Module:** Direct plug-and-play integration with native vehicle telemetry and inertial sensors.

Technical Performance & Latency Metrics

- **Compute Efficiency:** Chameleon-inspired asymmetric monocular scanning handles low-risk monitoring, drastically cutting baseline processing power compared to full 3D point-cloud generation.
- **Sub-5ms Emergency Reflex:** A hardware-level Locust LGMD bypass loop detects imminent collision expansion directly on the edge, skipping typical 50–150ms software pipeline delays to trigger braking or evasive steering in under 5ms.

Strategic Risk & Insurance Feasibility

- **Accelerated Time-to-Market:** MATLAB/Simulink and dSPACE HIL pipelines cut software verification cycles and reduce expensive track testing iterations.
- **Lower Liability Risk:** Isolating the 5ms emergency reflex in a dedicated ASIL-D certified partition (ISO 26262) guarantees a fail-safe backup, minimizing manufacturer liability and warranty risks.

OEM Integration & Operational Feasibility

- **Zero Infrastructure Dependency:** Operates reliably on non-laned, unstructured roads using biomimetic vision and Theta* + DWA planners without requiring costly roadside smart infrastructure upgrades.
- **No Wiring Harness Redesigns:** Generates AUTOSAR Adaptive components via Embedded Coder that flash directly onto existing OEM domain controllers, eliminating assembly line or hardware overhaul costs.
- **Compute-Efficient Footprint:** Asymmetric monocular scanning cuts perception compute overhead, allowing advanced risk evaluation on compact, low-power chips without auxiliary thermal cooling systems.



IMPACT AND BENEFITS



Socio-Economic & Infrastructural Benefits

- **Unstructured Traffic Navigation:** Combines Theta* global pathfinding with local DWA to seamlessly navigate unlaned roads, unmarked surfaces, and chaotic mixed traffic (pedestrians, auto-rickshaws, and stray animals).
- **Zero Infrastructure Dependency:** Functions independently of smart roadside units, painted lanes, or external infrastructure, allowing immediate deployment across rural, semi-urban, and unpaved roads.
- **Significant Fatal Crash Reduction:** Directly targets high-risk collision scenarios (sudden cut-ins and pedestrian step-outs) using sub-5ms hardware safety overrides to curb India's fatal road accidents (1.7+ lakh annual deaths).

Alignment with National Schemes & Social Impact

- **National Road Safety Mission (MoRTH):** Supports India's mandate to cut road fatalities by 50% using low-cost, indigenous active safety technology.
- **PM Gati Shakti & Logistics Efficiency:** Improves freight safety and minimizes transit delays caused by accidents or severe weather (monsoons and heavy fog).
- **Make in India & Digital India:** Replaces imported 3D LiDAR setups with local COTS hardware and native AUTOSAR code, building domestic IP in autonomous software.

Direct Impact on Target Audience & Everyday Citizens

- **Democratizing Level 2+ ADAS:** Brings sensor BOM under ₹1 Lakh for cost-sensitive A/B-segment passenger cars and commercial fleets.
- **Tackling Indian Driving Dynamics:** Deploys biomimetic asymmetric monocular vision, Snell's Law refraction matrices for water-filled pothole depth estimation, and sub-5ms Locust LGMD emergency bypass loops to prevent sudden cut-in collisions.



RESEARCH AND RESOURCES



Government and Policy Research-

MoRTH Road Accidents in India Report: Published by the Ministry of Road Transport and Highways, providing official statistical datasets on crash severity and fatalities across Indian roads.

<https://www.dataforindia.com/road-accident-deaths/>

MoRTH Official Portal: Official ministry documentation detailing Indian transport infrastructure, safety regulations, and vehicle compliance guidelines.

<https://morth.gov.in/#/>

Commercial Software and HIL Hardware Tooling

MathWorks Automated Driving Toolbox: Technical documentation for multi-sensor perception, synthetic visual scanning, and path generation.

MathWorks Model Predictive Control (MPC) Toolbox: Official product documentation for receding-horizon trajectory tracking and dynamic vehicle control.

Biomimetic and Robotics Research Publications

Locust LGMD Looming Neural Model (PubMed): Research paper detailing the mathematical formulations of the Lobula Giant Movement Detector (LGMD) for visual looming triggers.

Locust LGMD Robot Collision System (ResearchGate): Engineering paper detailing hardware/FPGA implementation of LGMD-inspired neural bypass loops for mobile robots.

Path Planning and Swarm Intelligence Publications

Multi-Agent Ant Colony Optimization for Traffic Rerouting (PMC / NIH): Biomimetic research using Ant Colony Optimization (ACO) for dynamic vehicle rerouting and traffic coordination.

Swarm Intelligence in Autonomous Vehicle Navigation (IEEE Xplore): Engineering publications detailing decentralized ACO metaheuristics over vehicle mesh networks.